

ABHIJIT MANOHAR MANE

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PROFILE

C++ Software Developer and Graphics Programmer with 1.5+ years of professional experience in C/C++ software development, specializing in high-performance and mission-critical applications. Experienced in memory management, STL, data-cache optimization, profiling, debugging, unit testing, and performance optimization. Strong hands-on expertise in real-time rendering and graphics programming using C++, OpenGL, OpenGL ES, WebGL, GLSL, and DirectX, with experience developing graphics engines, rendering pipelines, shader-based effects, and COM-based graphics frameworks.

TECHNICAL SKILLS

Core Programming: C++, C, Java, Data Structures, Object-Oriented Programming

Graphics Programming: OpenGL 4.6, OpenGL ES, WebGL, DirectX 11

Shader Programming: GLSL, HLSL, Programmable Pipeline, Fixed-Function Pipeline

Rendering: Real-Time Rendering, 3D Graphics, PBR, Lighting, Shadow Mapping, FBO, Texture Mapping

GPU Computing: CUDA, OpenCL

System Programming: Win32 SDK, COM, X11, Android SDK

Performance: Memory Management, Performance Optimization

Tools: Git, Visual Studio, Vim

PROFESSIONAL EXPERIENCE

LiteCode Software Pvt. Ltd.

Pune, India

Software Developer

Nov 2023 – Apr 2025

- Developed and optimized mission-critical financial policy modules using C/C++, translating complex business and technical specifications into efficient software components.
- Applied low-level memory management and data-cache optimization techniques to improve application efficiency and runtime performance.
- Analyzed and optimized existing C/C++ code to improve execution efficiency, memory utilization, and overall system performance.
- Worked with complex business logic and software requirements to develop reliable and maintainable production modules.
- Debugged and resolved software issues through systematic code analysis, debugging, and performance investigation.
- Collaborated with development teams to understand requirements, implement enhancements, and maintain existing software components.

PROJECTS

F1 – Real-Time Racing Graphics Demo

[YouTube]

Tech Stack: C++, OpenGL 4.6, GLSL, Win32, Assimp, Framebuffer Objects, PBR, Shadow Mapping, Particle Systems, Cube Mapping, Depth of Field, Motion Blur, Bezier Camera, Font Rendering, Video Rendering

- Developed a real-time F1 racing scene demonstrating advanced rendering techniques and GPU-based graphics effects.
- Implemented PBR materials, shadow mapping, environment mapping, particle-based smoke, depth of field, motion blur, and framebuffer-based rendering.
- Integrated model loading, camera animation, lighting, texture mapping, and video/font rendering into an interactive OpenGL application.

HOPE – OpenGL Animated Demo

[YouTube]

Tech Stack: C++, OpenGL 4.6, GLSL, Win32, PBR, Shadow Mapping, Framebuffer Objects, Particle Systems, Alpha Blending, Environment Mapping, Video Rendering, Bezier Camera

- Developed an animated 3D real-time rendering application using OpenGL and GLSL.
- Implemented shadow mapping, particle systems, environment mapping, alpha blending, framebuffer rendering, video rendering, and Bezier camera animation.

GLInsight – Real-Time Rendering Engine

[YouTube]

Tech Stack: C++, OpenGL 4.6, GLSL, Win32 SDK, GDI/GDI+, COM, DLL, Assimp, FBO, Shader Pipeline, Camera Systems

- Developed a modular real-time rendering engine using OpenGL and COM-based components for graphics and UI functionality.
- Implemented reusable rendering components including shader management, framebuffer objects, textures, camera systems, MVP transformations, and model loading.
- Developed rendering effects including terrain, water, atmospheric scattering, instancing, particles, and environment mapping.

Khalish – WebGL Animated Demo

[YouTube]

Tech Stack: JavaScript, WebGL 2.0, GLSL, Particle Systems, Lighting, Instancing, Terrain Rendering, Transparency

- Developed an interactive WebGL 2.0 application demonstrating programmable shaders and real-time rendering techniques.
- Implemented particle effects, terrain rendering, lighting, instancing, and transparency.

Android OpenGL ES Application

[YouTube]

Tech Stack: Java, Android SDK, OpenGL ES 3.2, GPU Rendering, 2D/3D Graphics, Texture Sampling, FBO, Render-to-Texture

- Developed an Android OpenGL ES 3.2 application featuring texture sampling, framebuffer-based rendering, render-to-texture, and interactive 2D/3D graphics.

Undarachi Topi – 3D OpenGL Animation Demo

[YouTube]

Tech Stack: C, OpenGL Fixed-Function Pipeline, X11, Linux, Geometric Primitives, Mathematical Curves, Terrain, Water, Particle Systems

- Developed a 3D animation demo using OpenGL on Linux with the X11 API.
- Implemented terrain and water rendering, particle systems, geometric primitives, mathematical curves, and scene transitions.

Image Editor (GUI Desktop Application)

[YouTube]

Tech Stack: C++, Win32 SDK, GDI, COM (Class Factory, Containment, Aggregation), Bitmap Manipulation, Pixel Filtering (Sepia, Negative, Desaturation)

- Developed a native Win32 bitmap image editor featuring advanced pixel manipulation filters (Sepia, Negative, Desaturation).
- Built image processing components using COM Class Factory, Containment, and Aggregation structural patterns.

Snake Game & Win32 Desktop Demos

[YouTube]

Tech Stack: C++, Win32 SDK, GDI, COM SAPI, Low-Latency Audio Feedback, Keyboard Input Handling, Timer-Based 2D Animation

- Developed a Win32 Snake game with keyboard controls and interactive, low-latency audio feedback implemented via COM SAPI components.

CERTIFICATIONS

- **AMC Certified Developer – Real-Time Rendering (RTR6) 2024** – AstroMediComp | Grade: 10.0/10.0 | Jun 2026
- **Data Structures Course Certification** – Core2web Technologies (Nov 2022)

EDUCATION

Dr. Babasaheb Ambedkar Technological University (DBATU)

Bachelor of Technology in Computer Engineering | CGPA: 7.83

Maharashtra, India

Aug 2019 – Jul 2023